Reconciliation Report (expected output)

Deshpande Print Solutions — August 2026
Ledger file: **ledger_aug2026.pdf** (32 rows) Bank statement: **bank_statement_aug2026.pdf** (32 rows)
Run 1 tolerances: date window ±3 days, amount tolerance 100 (absolute). Description threshold 80% (rapidfuzz token_set_ratio, case-insensitive).

Matched	Partial	Unmatched	Total
24	2	12	38

Of the 24 matched, 5 matched with tolerance applied (tolerance flag = true). Total = matched pairs + partial pairs + unmatched rows from both files.

Rows are sorted by date. Green = Matched (exact or with tolerance), yellow = Partial, red = Missing.

#	Date	Ledger particulars	Ledger (INR)	Bank (INR)	Bank narration	Status	Reason	Sim %
1	2026-08-01	Rent - Shah Properties - Aug 2026	25,000.00	25,000.00	NEFT SHAH PROPERTIES RENT AUG 2026	Matched (tolerance)	Matched with tolerance (date +2d)	100
2	2026-08-03	Receipt - Kalpana Events - Inv 0227	18,500.00	18,450.00	NEFT CR KALPANA EVENTS INV 0227	Matched (tolerance)	Matched with tolerance (amount -50.00)	87
3	2026-08-03	Meera Paper Mart - Bill 1187	42,300.00	42,300.00	CLG CHQ 000214 MEERA PAPER MART BILL 1187	Matched (tolerance)	Matched with tolerance (date +3d)	100
4	2026-08-04	Electricity bill - Adani - Jul 2026	6,842.00	6,842.00	BILLPAY ADANI ELECTRICITY JUL 2026	Matched	Exact match	95
5	2026-08-04	Internet - Jio Fiber - Aug 2026	1,179.00	1,179.00	BILLPAY JIO FIBER INTERNET AUG 2026	Matched	Exact match	100
6	2026-08-04	—	—	17.70	SMS ALERT CHARGES JUL 2026	Missing	Missing from ledger	—
7	2026-08-05	Receipt - Bright Minds Academy - Inv 0229	32,000.00	32,000.00	IMPS CR BRIGHT MINDS ACADEMY INV 0229	Matched	Exact match	89
8	2026-08-06	Ganesh Inks - Toner CMYK set	15,240.00	—	—	Missing	Missing from bank	—
9	2026-08-06	—	—	12,540.00	NEFT GANESH INKS TONER CMYK SET	Missing	Missing from ledger	—
10	2026-08-07	Salary - Sunil Pawar - Jul 2026	22,000.00	22,000.00	NEFT SALARY SUNIL PAWAR JUL 2026	Matched	Exact match	100
11	2026-08-07	Salary - Priya Naik - Jul 2026	19,500.00	19,500.00	NEFT SALARY PRIYA NAIK JUL 2026	Matched	Exact match	100
12	2026-08-07	Salary - Arjun Mhatre - Jul 2026	16,000.00	16,000.00	NEFT SALARY ARJUN MHATRE JUL 2026	Matched	Exact match	100
13	2026-08-08	Courier - DTDC - client dispatch	1,240.00	1,240.00	UPI DTDC COURIER CLIENT DISPATCH	Matched	Exact match	100
14	2026-08-10	Sundry expenses	2,360.00	2,360.00	UPI AMAZON PAY ORDER 4471	Partial	Description similarity 25% < 80%	25
15	2026-08-11	Receipt - Nirvana Wellness Spa - Inv 0230	12,500.00	—	—	Missing	Missing from bank	—
16	2026-08-12	Advance - Meera Paper Mart	5,000.00	5,000.00	NEFT MEERA PAPER MART ADVANCE	Matched	Exact match	100
17	2026-08-12	Advance - Ganesh Inks	5,000.00	5,000.00	NEFT GANESH INKS ADVANCE	Matched	Exact match	100
18	2026-08-13	Machine AMC - Sai Copier Services - Q2	9,000.00	9,000.00	NEFT SAI COPIER SERVICES AMC Q2	Matched	Exact match	91

#	Date	Ledger particulars	Ledger (INR)	Bank (INR)	Bank narration	Status	Reason	Sim %
19	2026-08-14	Receipt - Kalpana Events - Inv 0231	27,600.00	27,600.00	NEFT CR KALPANA EVENTS INV 0231	Matched	Exact match	87
20	2026-08-14	Packaging - Om Sai Packers	4,900.00	4,810.00	UPI OM SAI PACKERS PACKAGING	Matched (tolerance)	Matched with tolerance (amount -90.00)	100
21	2026-08-17	Fuel - delivery van	3,200.00	3,200.00	UPI HP PETROL PUMP FUEL DELIVERY VAN	Matched	Exact match	100
22	2026-08-17	—	—	12,500.00	NEFT CR NIRVANA WELLNESS SPA INV 0230	Missing	Missing from ledger	—
23	2026-08-18	GST payment - Jul 2026	21,870.00	21,870.00	GST PAYMENT JUL 2026 CPIN 26081812340	Matched	Exact match	100
24	2026-08-18	—	—	7,500.00	NEFT CR RUSHABH TRADERS	Missing	Missing from ledger	—
25	2026-08-19	Receipt - Bright Minds Academy - Inv 0233	1,25,000.00	—	—	Missing	Missing from bank	—
26	2026-08-19	—	—	1,24,400.00	NEFT CR BRIGHT MINDS ACADEMY INV 0233	Missing	Missing from ledger	—
27	2026-08-20	Meera Paper Mart - Bill 1203	38,750.00	38,750.00	NEFT MEERA PAPER MART BILL 1203	Matched	Duplicate record	100
28	2026-08-20	Meera Paper Mart - Bill 1203	38,750.00	—	—	Missing	Missing from bank; Duplicate record	—
29	2026-08-21	Loan EMI - Girnar Bank - Aug 2026	11,430.00	11,430.00	ACH DR GIRNAR BANK TERM LOAN EMI AUG 2026	Matched	Exact match	100
30	2026-08-22	Receipt - Nirvana Wellness Spa - Chq 118834	9,800.00	9,800.00	CLG CR NIRVANA WELLNESS SPA CHQ 118834	Matched (tolerance)	Matched with tolerance (date +2d)	90
31	2026-08-24	Petty cash withdrawal	5,000.00	5,000.00	ATM CASH WDL DADAR WEST	Partial	Description similarity 50% < 80%	50
32	2026-08-25	Stationery - Kamal Stores	1,875.00	1,875.00	UPI KAMAL STORES STATIONERY	Matched	Exact match	100
33	2026-08-26	Receipt - Prime Realty - Inv 0235	45,000.00	—	—	Missing	Missing from bank	—
34	2026-08-27	Ganesh Inks - Bill 4470	8,320.00	8,320.00	NEFT GANESH INKS BILL 4470	Matched	Exact match	100
35	2026-08-28	Sai Copier Services - Spare parts	6,600.00	—	—	Missing	Missing from bank	—
36	2026-08-29	Salary advance - Arjun Mhatre	5,000.00	5,000.00	IMPS ARJUN MHATRE SALARY ADVANCE	Matched	Exact match	100
37	2026-08-31	Website hosting - Hostinger - annual	3,588.00	3,588.00	CARD HOSTINGER WEBSITE HOSTING ANNUAL	Matched	Exact match	100
38	2026-08-31	—	—	295.00	MONTHLY MAINT CHARGES AUG 2026	Missing	Missing from ledger	—

Appendix A — What the non-trivial rows test

Ledger row	Bank row	Behaviour under test	Success criterion
01/08 Rent Shah Properties 25,000	03-Aug 25,000	Date tolerance +2 days (recorded Sat, posted Mon)	Tolerance, Automatic Matching
03/08 Meera Paper Mart Bill 1187 42,300	06-Aug 42,300	Date tolerance boundary: exactly +3 days (must be inclusive)	Tolerance
03/08 Kalpana Events Inv 0227 18,500	03-Aug 18,450	Amount tolerance -50 (within 100)	Tolerance
14/08 Om Sai Packers 4,900	14-Aug 4,810	Amount tolerance -90 (within 100 absolute; fails 1% in Run 2)	Tolerance
19/08 Bright Minds Inv 0233 1,25,000	19-Aug 1,24,400	Amount diff 600: outside 100 absolute (both flagged), inside 1% in Run 2	Tolerance, Flagging
06/08 Ganesh Inks toner 15,240	06-Aug 12,540	Transposition error, same date and description: must NOT match	Flagging
11/08 Nirvana Inv 0230 12,500	17-Aug 12,500	Same amount and description but 6 days apart: outside window, both flagged	Flagging
12/08 Advance Meera 5,000 / Advance Ganesh 5,000	12-Aug Ganesh 5,000 / Meera 5,000 (opposite order)	Two candidates pass date+amount; each must pair with the best description, not the first candidate	Automatic Matching
10/08 Sundry expenses 2,360	10-Aug UPI AMAZON PAY 2,360	Date and amount pass, description 25%: Partial	Automatic Matching
24/08 Petty cash withdrawal 5,000	24-Aug ATM CASH WDL 5,000	Date and amount pass, description 50%: Partial	Automatic Matching
20/08 Meera Paper Mart Bill 1203 38,750 (x2)	20-Aug 38,750 (x1)	Duplicate ledger entry: both copies flagged Duplicate; second copy also Missing from bank	Flagging
26/08 Prime Realty Inv 0235 45,000; 28/08 Sai Copier spares 6,600	(none)	Ledger entries not yet in bank (cheque not cleared): Missing from bank	Flagging
(none)	04-Aug SMS charges 17.70; 18-Aug Rushabh Traders 7,500; 31-Aug maint charges 295	Bank entries never recorded in ledger: Missing from ledger	Flagging
Headers: Date / Voucher No. / Particulars / Amount (INR)	Headers: Txn Date / Ref No. / Description / Amount (INR) / Dr/Cr / Balance (INR)	Column mapping must pick date, description and amount from differently named headers and ignore extra columns	Automatic Matching
Dates DD/MM/YYYY	Dates DD-Mon-YYYY	Date normalisation to YYYY-MM-DD from two formats; DD/MM must not be read as MM/DD	Automatic Matching
Amounts 1,25,000.00	Amounts 1,24,400.00	Indian digit grouping: commas must be stripped before float conversion	Automatic Matching

Appendix B — Same input files, different tolerance settings

Run 2: date ±3 days, amount tolerance 1% (percentage, of the ledger amount). Summary: Matched 24, Partial 2, Unmatched 12, Total 38. Counts are identical to Run 1, but the rows differ: the Om Sai Packers pair (diff 90 = 1.84%) splits into two Missing rows, and the Bright Minds Inv 0233 pair (diff 600 = 0.48%) becomes a tolerance match:

#	Date	Ledger particulars	Ledger (INR)	Bank (INR)	Bank narration	Status	Reason	Sim %
1	2026-08-14	Packaging - Om Sai Packers	4,900.00	—	—	Missing	Missing from bank	—
2	2026-08-14	—	—	4,810.00	UPI OM SAI PACKERS PACKAGING	Missing	Missing from ledger	—
3	2026-08-19	Receipt - Bright Minds Academy - Inv 0233	1,25,000.00	1,24,400.00	NEFT CR BRIGHT MINDS ACADEMY INV 0233	Matched (tolerance)	Matched with tolerance (amount -600.00)	89

Run 3: date window 0 days, amount tolerance 0 (exact matches only). Summary: Matched 19, Partial 2, Unmatched 22, Total 43. The five tolerance matches of Run 1 split into ten Missing rows; the two Partial rows are unchanged because their date and amount were exact:

#	Date	Ledger particulars	Ledger (INR)	Bank (INR)	Bank narration	Status	Reason	Sim %
1	2026-08-01	Rent - Shah Properties - Aug 2026	25,000.00	—	—	Missing	Missing from bank	—
2	2026-08-03	Receipt - Kalpana Events - Inv 0227	18,500.00	—	—	Missing	Missing from bank	—
3	2026-08-03	Meera Paper Mart - Bill 1187	42,300.00	—	—	Missing	Missing from bank	—
4	2026-08-03	—	—	25,000.00	NEFT SHAH PROPERTIES RENT AUG 2026	Missing	Missing from ledger	—
5	2026-08-03	—	—	18,450.00	NEFT CR KALPANA EVENTS INV 0227	Missing	Missing from ledger	—
6	2026-08-06	—	—	42,300.00	CLG CHQ 000214 MEERA PAPER MART BILL 1187	Missing	Missing from ledger	—
7	2026-08-14	Packaging - Om Sai Packers	4,900.00	—	—	Missing	Missing from bank	—
8	2026-08-14	—	—	4,810.00	UPI OM SAI PACKERS PACKAGING	Missing	Missing from ledger	—
9	2026-08-22	Receipt - Nirvana Wellness Spa - Chq 118834	9,800.00	—	—	Missing	Missing from bank	—
10	2026-08-24	—	—	9,800.00	CLG CR NIRVANA WELLNESS SPA CHQ 118834	Missing	Missing from ledger	—

Appendix C — Assumptions used to derive this expected output

- One-to-one matching: once a bank row is matched to a ledger row it is not available to other ledger rows.
- When several bank rows pass the date and amount checks, the one with the highest description similarity is chosen.
- Tolerances are inclusive (a 3-day gap passes a ±3-day window; a difference of exactly 100 would pass an absolute tolerance of 100).
- Percentage tolerance is computed on the ledger amount.
- Description similarity = rapidfuzz fuzz.token_set_ratio with default_process (case-folded, punctuation stripped). Score ≥ 80 = Matched, otherwise Partial.
- Status follows the Criterion C flowchart: Partial means date and amount passed but description scored below 80%. Matches that needed tolerance are Matched with tolerance_applied = true.
- Duplicate detection is on (date, description, amount) within a file; every copy in the group gets the Duplicate record reason.
- Amounts are compared ignoring Dr/Cr direction, as the design maps only date, description and amount.